Applications

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Edge-based imitation learning for autonomous tasks: a case study on tractor-trailer positioning

Edge-basiertes Imitationslernen für autonome Aufgaben: Eine Fallstudie anhand der Traktor-Anhänger-Positionierung

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Abstract: In this work, we investigate imitation learning and offline reinforcement learning approaches for controlling a tractor-trailer system with a steerable front axle during backward maneuvers. We first develop an expert policy using a kinematic model and Proximal Policy Optimization to generate high-quality demonstration data. Leveraging this dataset, we compare Behavioral Cloning, Implicit Q-Learning, and TD3-BC as representative offline learning methods. Our experiments evaluate sample efficiency, generalization to unseen initial conditions, and model parameter count highlighting trade-offs between pure imitation, value-based optimization, and model complexity. We further analyze trajectory-level behavior and robustness under varying starting configurations. Results show that the applied methods can achieve performance comparable to or exceeding the expert policy while requiring fewer demonstrations, providing insights into the design of data-efficient learning-based controllers for complex trailer systems.

Keywords: tractor-trailer control; imitation learning; offline reinforcement learning; learning from demonstrations

Zusammenfassung: In dieser Arbeit untersuchen wir Methoden des Imitationslernens und des Offline-Reinforcement-Learnings zur Steuerung eines Traktor-Anhänger-Gespanns mit Drehschemel bei Rückwärtsfahrmanövern. Zunächst entwickeln wir eine Expert Policy auf Basis eines kinematischen Modells und Proximal Policy Optimization, um hochwertige Demonstrationsdaten zu erzeugen. Auf Grundlage dieses Datensatzes vergleichen wir Behavioral Cloning, Implicit Q-Learning und TD3-BC als repräsentative Offline-Lernmethoden. Unsere Experimente bewerten die Sample-Effizienz, die Generalisierung auf unbekannte Startkonfigurationen sowie die Anzahl der Modellparameter und beleuchten die Trade-offs zwischen reinem Imitationslernen, wertbasierter Optimierung und Modellkomplexität. Darüber hinaus analysieren wir das Verhalten auf Trajektorienebene und die Robustheit unter variierenden Anfangsbedingungen. Die Ergebnisse zeigen, dass die eingesetzten Methoden eine vergleichbare oder bessere Leistung als die Expert Policy erreichen können, während sie weniger Demonstrationen benötigen, und liefern wertvolle Erkenntnisse für die daten-effiziente Gestaltung lernbasierter Regler für komplexe Anhängersysteme.

Schlagwörter: Traktor-Anhänger-Positionierung; Imitationslernen; Offline Reinforcement Learning; Learning from Demonstrations

1 Introduction

The automation of repetitive processes in spatially constrained environments offers significant potential for efficiency gains. Imitation learning (IL) enables the development of effective control policies from expert demonstrations, particularly in scenarios where complex dynamics or environmental limitations challenge traditional methods. Key concerns include sample efficiency, limited computational resources, and restricted sensor availability, alongside the need for generalizability and robustness.

Recent advances in robotics have shifted control strategies from fixed algorithms to adaptive, learning-based methods capable of handling dynamic conditions. In agriculture, such methods address the variability of products

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and field conditions, which differs from the predictability of industrial settings. Tasks such as planting, harvesting, or maneuvering equipment require adaptability to changes in product size, appearance, and terrain [1], [2].

One such challenge is the reverse positioning of drawbar trailers with steerable front axles – a common agricultural operation requiring precise alignment in confined spaces and under varying soil or weather conditions. IL is particularly suited to this problem, as it allows autonomous systems to learn expert-level positioning strategies by observing skilled operators. Leveraging these demonstrations enables automation of such tasks while retaining adaptability to environmental feedback.

The primary objective of this work is to investigate and compare imitation learning methods for autonomous trailer positioning. To systematically evaluate these methods before deploying them on a physical system, we employ a simulation-based approach: a virtual expert policy is first trained via reinforcement learning (RL) in a simulation environment, which then serves as a source of diverse, consistent demonstration trajectories. This approach enables controlled experimentation with imitation learning algorithms – including variations in dataset size, observation noise, and model capacity – without the risks, costs, and logistical constraints associated with real-world data collection.

While simulation-based training cannot fully replace field experiments, it provides a valuable preliminary investigation to identify promising IL methods before deployment on physical systems. Taken together, this simulation-based setup provides the foundation for imitation learning methods that can efficiently learn control policies from limited demonstrations, addressing practical constraints of edge computing environments where computational and sensory resources are limited.

2 Related work

While tractor-trailer backing has been extensively studied for simpler configurations such as single-axle, non-steerable trailers, the problem remains considerably more challenging for articulated drawbar trailers with a steerable front axle. These systems exhibit more complex kinematic behavior and a higher risk of instability, making precise control difficult, especially in confined environments. The task is further complicated by nonlinear dynamics and nonholonomic constraints [3].

Early works addressed this challenge for simpler configurations through kinematic modeling and classical control methods, proposing motion-planning approaches based on geometric constraints to achieve stable backward motion [4]–[6]. Subsequent research introduced cascaded control schemes for path tracking and stabilization of reversing general two-trailer configurations, demonstrating robust performance in complex maneuvers [7]. This was later extended by comprehensive frameworks that combine path planning and path-following control for two-trailer systems with car-like tractors [8]. More recent work employed model predictive control (MPC) to integrate kinematic models with optimization techniques, achieving precise backing control for non-steerable trailers [9] or path following for different trailer configurations [10].

Building on this, recent advancements increasingly leverage learning based methods, particularly Deep RL, to address the complexity of articulated trailers. For instance, a Deep RL approach combined with fuzzy logic has been proposed for backing up a self-driving truck-trailer vehicle, achieving precise control in simulated environments [11]. This method leverages Deep RL to handle the nonlinear dynamics of the system while using fuzzy logic to improve robustness. Similarly, a policy gradient method has been developed to train a tractor-trailer mobile robot system for backing [12]. However, most works focus on single-axle trailers, and RL often requires extensive exploration and a well-defined reward function, which can be difficult to design for such tasks. These challenges motivate the use of imitation learning, where policies are learned directly from expert demonstrations, bypassing the need for explicit reward design and enabling more efficient and practical training for complex trailer configurations.

Imitation learning (IL) has been widely studied, including classical methods such as Behavioral Cloning (BC) [13], DAgger [14], and GAIL/IRL [15], [16]. BC directly maps states to expert actions and is simple and sample-efficient, but prone to compounding errors. DAgger improves robustness via iterative expert queries, while GAIL/IRL infer reward functions to generalize better, though at increased training complexity. Recent IL advances incorporate recurrent policies, attention, or multi-task learning to enhance generalization.

Offline reinforcement learning (offline RL) can leverage demonstration datasets, blending imitation with value-based learning [17]. Methods such as Implicit Q-Learning (IQL) [18] and TD3-BC [19] integrate BC with offline RL objectives, improving robustness to limited or noisy demonstrations and exploiting reward signals when available. While IL focuses on direct behavior cloning, offline RL better handles suboptimal data at the cost of increased tuning and potential extrapolation errors [17].

We evaluate BC, IQL, and TD3-BC on trailer backing with steerable front axles, varying demonstration quantity, observation noise, and network size to quantify sample efficiency, robustness, and generalization.

3 Methods

3.1 Expert policy simulation

To generate demonstrations for IL and offline RL, we first develop an expert policy using reinforcement learning. While model-based approaches such as MPC offer precise trajectory tracking when accurate kinematic models are available, learning-based methods provide distinct advantages for this task: they require less manual parameter tuning and enable flexible extension to obstacle avoidance or multiple goal configurations through reward function design. To the authors' knowledge, studies of learning-based methods for steerable-axle trailer configurations remain limited, motivating the present investigation.

The tractor–trailer system is modeled kinematically, with state variables describing the positions and orientations of the tractor and trailer units, as well as articulation and steering angles (Figure 1). This kinematic formulation is appropriate for low-speed maneuvers typical in precision agriculture and trailer positioning tasks. At such speeds, longitudinal velocity variations are minimal, and the system dynamics are dominated by geometric constraints rather than inertial effects. Consequently, the kinematic model – which explicitly models geometric relationships without velocity or acceleration terms – provides a sufficient representation for learning control policies [4], [10]. This simplification reduces model complexity while retaining the essential nonholonomic constraints that govern the system's behavior.

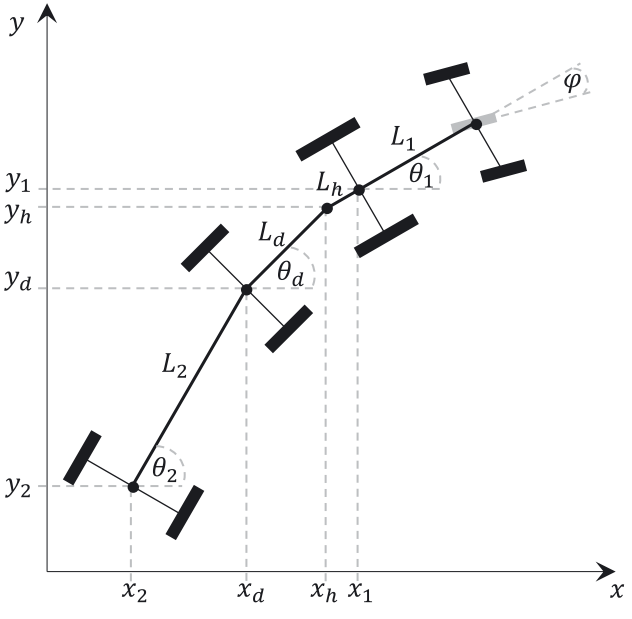


Figure 1: Model of a tractor–trailer system with steerable trailer axle, defining key state variables and geometric parameters.

The position and orientation of the tractor (x_1, y_1, θ_1) are defined at the center of the rear axle, with L_1 denoting the wheelbase and φ the steering angle. The hitch point is located at a distance L_h behind the rear axle:

$$x_h = x_1 - L_h \cos\theta_1$$
$$y_h = y_1 - L_h \sin\theta_1$$

The drawbar of length L_d and orientation θ_d connects the hitch point with the front axle of the trailer:

$$x_d = x_h - L_d \cos\theta_d$$
$$y_d = y_h - L_d \sin\theta_d$$

The trailer position and orientation (x_2, y_2, θ_2) are defined at the center of the rear axle, where L_2 is the trailer wheelbase:

$$x_2 = x_d - L_2 \cos\theta_2$$
$$y_2 = y_d - L_2 \sin\theta_2$$

The state vector of the system is given by:

$$\underline{x} = \left[x_1, y_1, \theta_1, \theta_d, \theta_2\right]^T \tag{1}$$

The corresponding state equations are:

$$\begin{aligned}
\dot{x}_1 &= v_1 \cos\theta_1 \\
\dot{y}_1 &= v_1 \sin\theta_1 \\
\dot{\theta}_1 &= \frac{v_1}{L_1} \tan\varphi \\
\dot{\theta}_d &= \frac{v_{hy} \cos\theta_d - v_{hx} \sin\theta_d}{L_d} \\
\dot{\theta}_2 &= \frac{v_{\mathrm{d}y} \cos\theta_2 - v_{\mathrm{d}x} \sin\theta_2}{L_2},
\end{aligned} \tag{2}$$

where (v_{hx}, v_{hy}) and $(v_{\mathrm{d}x}, v_{\mathrm{d}y})$ are the velocity components in x- and y-direction of the hitch point and the trailer front axle, derived from the equations above. Unlike standard truck–semitrailer kinematic models, e.g. [20], the steerable drawbar trailer adds an extra degree of freedom via θ_d, requiring state variables to capture the articulation and trailer steering [4]. This difference in kinematic structure leads to increased nonholonomy and tighter stability margins during reversing maneuvers.

Note that the trailer position (x_2, y_2) is algebraically determined by the state variables (Eq. (1)) through a kinematic chain. Thus, it is not necessary to include (x_2, y_2) as explicit state variables for the dynamics, which depend only on the angles $(\theta_1, \theta_d, \theta_2)$ and the tractor position (x_1, y_1). However, the trailer position must be computed and monitored during execution to evaluate task success (i.e. goal reaching) and detect events like jackknifing.

Building upon that model, the control task can now be formulated as an optimization problem within a reinforcement learning framework. The aim is to design a policy that guides the tractor-trailer system from arbitrary initial states to a desired target configuration without collisions or jackknifing. To provide the learning algorithm with informative feedback, the reward function $r(\underline{x}, \varphi)$ combines a shaped continuous component r_c with a discrete sparse reward r_d and a time penalty r_t.

The continuous reward encourages smooth convergence toward the target state, while the sparse term penalizes the terminal trailer orientation error. The target is defined as a trailer position $\left(x_2^*, y_2^*\right)$ with orientation θ_2^*, which – without loss of generality – is transformed into the origin (0,0,0) of the coordinate frame.

The continuous component is given by

$$r_c = \frac{4}{1+d} - 3 \tag{3}$$

where the distance metric d combines lateral displacement and heading error of the trailer:

$$d = w_y \frac{|y_2|}{y_{\max}} + w_\theta \frac{|\theta_2|}{\theta_{\max}}. \tag{4}$$

The normalization constants $y_{\max}$ and $\theta_{\max}$ represent the maximum expected magnitudes of lateral displacement and heading error. The weighing factors w_y and w_θ are chosen as such that the metric d maps to approximately 1 for the largest observable values. Additionally a time penalty r_t per time step is subtracted to encourage efficient task completion and rewarding progress toward the goal.

When reaching the final state of an episode, defined as the trailer position satisfying $x_2 <= 0.0$ m, a sparse reward.

$$r_d = w_d \ \mathrm{Std}(\theta_1, \theta_d, \theta_2) \tag{5}$$

is subtracted from the total reward. This term penalizes the spread of the final articulation angles. Under the assumption that the angles remain within a continuous range without discontinuities at the $\pm\pi$ boundary during task execution, the standard deviation provides a rotation-invariant measure of angular dispersion.

When jackknifing occurs, which is defined as

$$\mathrm{Var}(\theta_1, \theta_d, \theta_2) \geq 0.4, \tag{6}$$

where the variance is calculated over the three angles θ_1, θ_d, and θ_2 under the same continuity assumption, the episode terminates with a reward of -1.0. This threshold-based penalty ensures that unstable configurations are immediately identified and penalized.

The resulting environment setup forms the basis for training an expert policy using Proximal Policy Optimization (PPO). PPO is chosen due to its robustness, sample efficiency, and ability to handle continuous action spaces without requiring complex value function tuning. In this context, PPO iteratively updates the policy network while respecting a trust region constraint, ensuring stable learning and avoiding destructive policy updates – an essential property given the non-linear, unstable dynamics of reversing with a steerable-axle trailer.

3.2 Imitation learning algorithms

In this work, three representative approaches for learning from demonstrations are compared:

BC serves as a supervised learning baseline [13], where the policy is trained to minimize the error between expert actions and predicted actions given the same states. BC does not require reward information and is simple to implement, but its performance strongly depends on the diversity and coverage of the demonstration dataset, limiting generalization to unseen situations.

IQL is an offline RL approach [18] that avoids explicit policy iteration. It learns a value function using expectile regression and derives the policy via advantage-weighted behavior cloning. Compared to quantile regression, expectile regression yields smoother gradients and more stable value estimates, which can reduce sensitivity to extreme or noisy returns in demonstration data. This makes IQL well suited for learning from imperfect expert trajectories, allowing it to leverage the structure of reinforcement learning while maintaining stability on demonstration-only datasets, even under sparse or noisy reward signals.

TD3-BC combines offline reinforcement learning with imitation learning [19]. It uses a critic to estimate Q-values and performs a weighted behavior cloning update on the actor. This hybrid approach can leverage reward information when available, improving robustness and performance compared to pure BC, while maintaining sample efficiency on limited demonstration datasets.

4 Experiments

4.1 Experimental setup

The experiments are conducted in a custom Gymnasium environment [21] modeling the kinematic tractor–trailer system with a steerable front axle on the trailer. The state space is defined according to equation (1). All positions are given in meters, and all angles in radians. The state vector is bounded by:

$$x_1 \in [-10.0, 40.0],$$
$$y_1 \in [-25.0, 25.0],$$
$$\theta_1, \theta_d, \theta_2 \in [-\pi, \pi].$$

The action space is one-dimensional and continuous, consisting solely of the steering angle of the tractor's front wheels:

$$\varphi \in \left[-\frac{\pi}{4}, \frac{\pi}{4}\right].$$

The longitudinal velocity is fixed to a constant backward speed.

At the start of each episode, the initial state is randomized within a bounded region to encourage generalization:

$$x_1 = 30.0 \pm 2.5 \text{ m},$$
$$y_1 = 0.0 \pm 2.5 \text{ m},$$
$$\theta_1 = 0.0 \pm \frac{\pi}{8},$$
$$\theta_2 = \theta_1 \pm \frac{\pi}{8},$$
$$\theta_d = \frac{\theta_1 + \theta_2}{2}.$$

The dynamics (Eqs. (2)) are simulated using forward Euler integration with a time step of $\Delta t = 0.05$ s. Each episode has a maximum length of 1,000 time steps but is terminated earlier upon goal achievement or jackknifing.

All experiments use the following parameter values: wheelbase tractor $L_1 = 3.6$ m, hitch distance $L_h = 1.0$ m, drawbar length $L_d = 3.0$ m, and trailer wheelbase $L_2 = 4.0$ m. These values are representative of mid-sized agricultural tractor-trailer combinations.

Reward function parameters are set to $w_y = 0.7$, $w_\theta = 0.3$, $y_{\max} = 25.0$ m, $\theta_{\max} = \pi$ rad, time penalty $r_t = 0.05$ per step, and sparse reward weight $w_d = 10.0$. These settings balance lateral alignment and heading accuracy while encouraging efficient task completion.

To enable imitation learning experiments, an expert policy is first trained using PPO in the described environment. This policy serves as a source of demonstration trajectories for IL, with the number of demonstrations varied across experiments (1, 2, 5, 10, 20, 50, and 100 trajectories). Each algorithm is then trained on these subsets, and performance is evaluated in the full environment from random initial states. This setup allows a direct comparison of sample efficiency and robustness with respect to the number of available demonstrations.

Figure 2 provides a block diagram illustrating the information flow in the system. Both the expert policy (trained via PPO) and the policies learned via imitation learning algorithms (BC, IQL, TD3-BC) receive as input the full state vector $\underline{x} = [x_1, y_1, \theta_1, \theta_d, \theta_2]^T$ and output a continuous steering angle $\varphi \in [-\pi/4, \pi/4]$. The longitudinal velocity is fixed to a constant backward speed throughout all experiments.

The expert policy generates demonstration trajectories by executing in the environment from randomized initial states, collecting state-action pairs for training the imitation learning algorithms. Each IL algorithm is then trained on subsets of these demonstrations, with performance evaluated based on cumulative reward, terminal state distributions, and success rates over multiple episodes.

4.2 Expert policy training

The virtual expert is trained in the simulation environment described in the previous section. The PPO parameters correspond to the default settings implemented in [22],

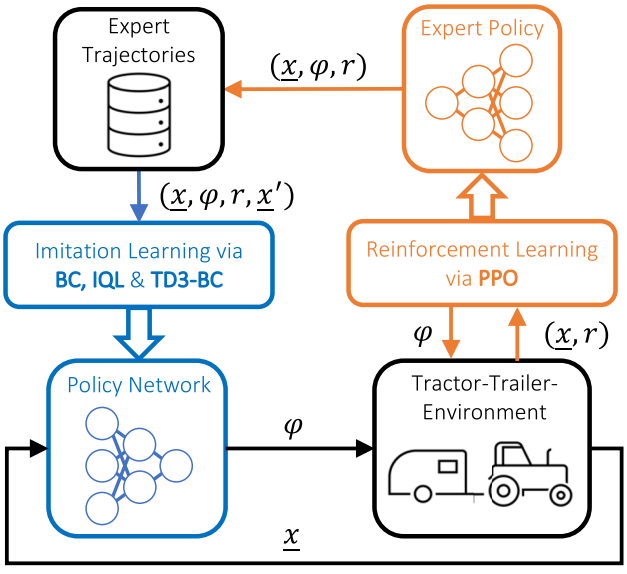


Figure 2: Block diagram of the expert policy and imitation learning algorithms: The truck-trailer environment is used to train an expert policy via PPO, which generates demonstration trajectories $(\underline{x}, \varphi, r)$. These demonstrations are then used to train policies via imitation learning algorithms. $\underline{x}'$ denotes the next state after applying action φ in state $\underline{x}$. The reward r consists of the components r_c, r_t and r_d as described in the previous section.

except for the following adjustments: the learning rate is set to 1×10^{-4}, and the clipping range is reduced to 0.01. A multilayer perceptron (MLP) with two hidden layers of 256 neurons each is used for both the policy and value networks. Training is performed for up to 10×10^6 time steps, with an early stopping callback that terminates training if no improvement is observed. During PPO training, after every 10 episodes, a validation step is performed to measure the average return over 20 rollouts. Training stops if no improvement is observed for 100 consecutive training episodes.

All experiments were conducted on a standard desktop computer (Apple Silicon Mac, 8-core CPU). Training the expert policy required approximately 2–3 h of wall-clock time, while training individual imitation learning models (BC, IQL, TD3-BC) on demonstration subsets typically completed within 5–15 min per configuration.

Results: The trained expert policy demonstrates stable convergence and high goal-reaching reliability, as illustrated in Figure 3. These trajectories serve as the benchmark and the training data source for the imitation learning methods evaluated later.

Figures 4 and 5 depict two representative trajectories of the expert policy. The first trajectory successfully reaches the target, whereas the second results in jackknifing. These examples highlight both the variability in expert demonstrations and the inherent challenges in controlling the tractor-trailer system. Overall, jackknifing occurs in approximately 2 % of the expert rollouts.

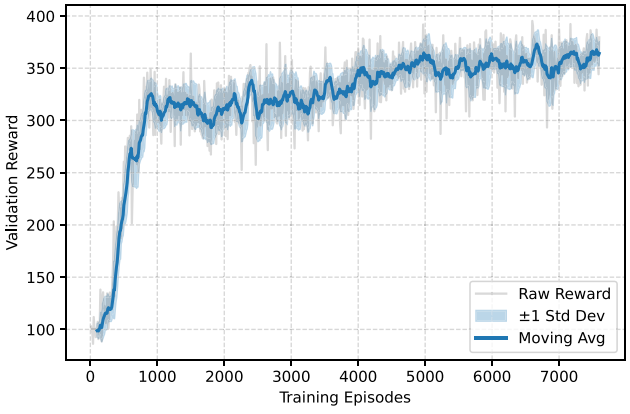


Figure 3: Average episodic return of the expert policy on the validation set: raw returns (gray) and smoothed moving average with standard deviation (blue) across training episodes.

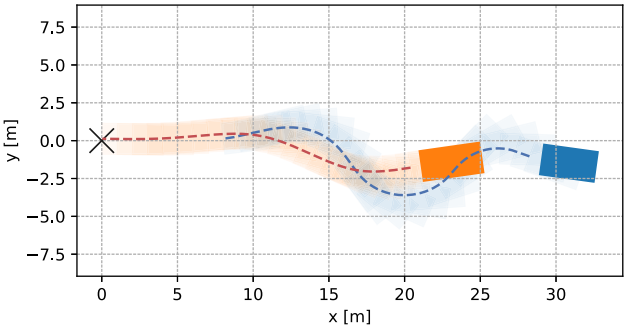


Figure 4: Example of an expert trajectory successfully reaching the goal. The blue and orange boxes indicate the initial positions of the tractor and the trailer, respectively. (Note: For visualization clarity, the boxes lengths represent the wheelbases of the tractor and trailer, while the actual vehicle lengths are longer due to overhangs not depicted here. The widths are set to 2.0 m for both boxes).

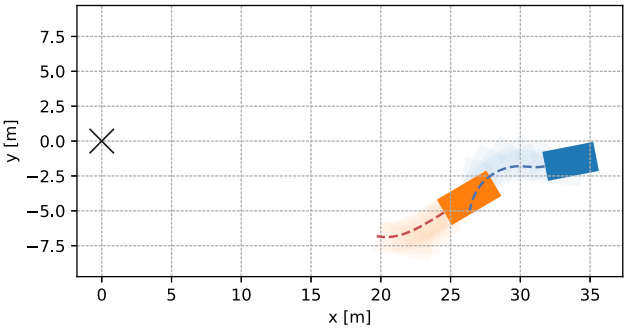


Figure 5: Example expert trajectory resulting in jackknifing.

4.3 Sample efficiency evaluation

To assess how efficiently different algorithms learn from demonstrations, BC, IQL, and TD3-BC are trained on datasets of varying size using [23]. Specifically, subsets of demonstrations containing 1, 2, 5, 10, 20, 50, and 100 trajectories are used. Performance is compared across these subsets to quantify sample efficiency, i.e., the ability to generalize from limited data. All algorithms are evaluated over 10×100 episodes per subset, measuring cumulative reward, terminal state distributions, and success rate. Terminal states are recorded at the end of each episode.

Results: The results provide a comprehensive view of each algorithm's performance as a function of the number of demonstration trajectories. Figure 6 illustrates the average episode return. All algorithms generally improve with more demonstrations, but notable differences in sample efficiency emerge. With very few demonstrations, both IQL and TD3-BC outperform BC. At 10 demonstrations, a small

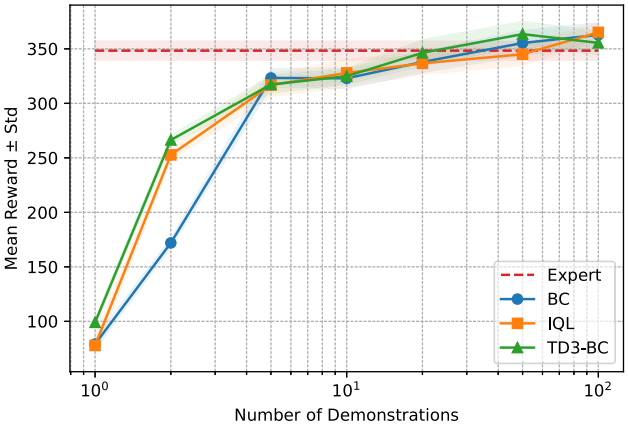


Figure 6: This figure shows the average episode return as a function of the number of demonstration trajectories for BC, IQL, and TD3-BC.

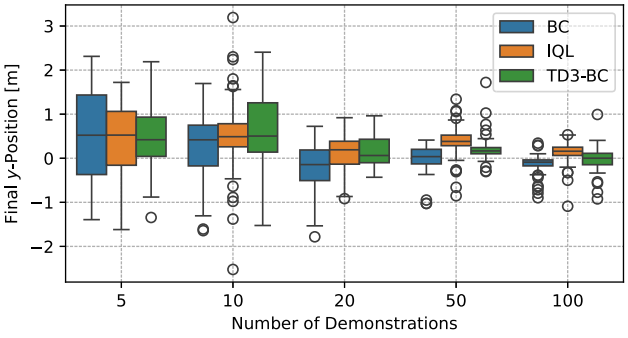


Figure 7: Trailer lateral position y_2 at the end of 100 episodes.

performance dip is visible. Interestingly, TD3-BC achieves a lower return with 100 demonstrations than with 50, whereas IQL shows steady improvement across the range of demonstrations.

To examine the quality of the final states more closely, Figures 7 and 8 depict the distributions of the trailer articulation angle θ_2 and lateral position y_2 at the end of each episode. As the number of demonstrations increases, the variability in both terminal state measures decreases, indicating more precise control and consistent task completion. These boxplots complement the reward analysis by highlighting how learning stability improves with larger datasets.

4.4 Robustness to observation noise

In this experiment, we replicate the setup of Experiment 1, but introduce noise to the observations during training. This scenario evaluates the resilience of the algorithms to imperfect or corrupted sensory inputs, a common challenge in real-world applications where, for example, an expert may rely on their own perceptual abilities (e.g., vision), while

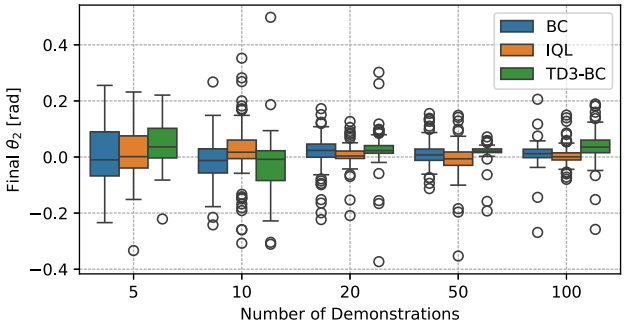


Figure 8: Trailer articulation angle θ_2 at the end of 100 episodes.

the observations used for training and imitation learning originate from other sensors such as cameras or GPS. Specifically, Gaussian noise with standard deviations $\sigma = [0.2, 0.2, 0.02, 0.02, 0.02]$ was added independently to each component of the state vector $(x_1, y_1, \theta_1, \theta_d, \theta_2)$ during training. This simulates sensor inaccuracies for position and orientation measurements.

Results: The results, shown in Figure 9, indicate how the average return on validation data evolves over episodes under noisy conditions. This allows comparison of algorithm robustness relative to the baseline scenario without observation noise. Across all settings, the returns remain below those achieved by the noise-free models, reflecting the detrimental impact of corrupted observations. It is important to note that observation noise was added exclusively during training; validation and evaluation were conducted on clean, noise-free state observations. This experimental design evaluates the robustness of training data quality rather than the impact of sensor noise during deployment. The advantage of the offline RL methods over BC observed with very few demonstrations is less pronounced in the noisy setting. Furthermore, all methods

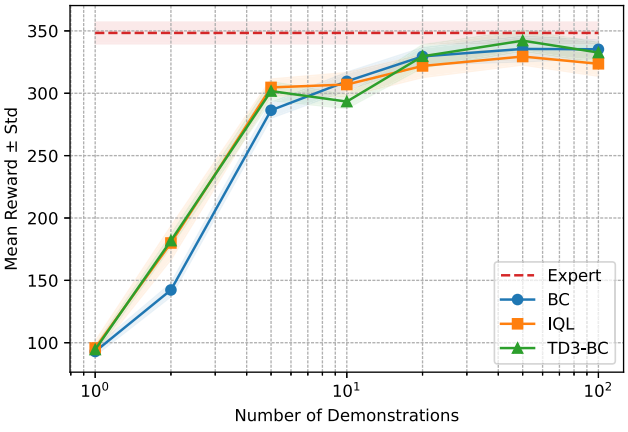


Figure 9: Validation return over episodes with observation noise ($\sigma = [0.2, 0.2, 0.02, 0.02, 0.02]$ for $(x_1, y_1, \theta_1, \theta_d, \theta_2)$).

exhibit a noticeable drop in performance at 100 demonstrations, suggesting that noise interacts with larger datasets in a way that limits the benefit of additional demonstrations.

4.5 Model capacity ablation

This experiment investigates the influence of model capacity on learning outcomes. Here, model capacity is defined as the number of learnable parameters in the network. Using 100 demonstration trajectories, the network size for the BC policy is systematically reduced, and the resulting performance is analyzed. This ablation study highlights the sensitivity of imitation learning to representational constraints and the trade-offs between model complexity and task performance.

Results: An examination of the validation returns reveals a consistent performance pattern: policies with smaller networks struggle to fully capture the expert behavior, while larger networks achieve higher returns (see Table 1). Specifically, the mean final returns increase from approximately 333 for 16 neurons to 371 for 256 neurons per hidden layer, demonstrating that increasing model capacity consistently improves task performance, though with diminishing gains beyond 128–256 neurons.

4.6 Discussion

The experiments are based on a set of expert demonstrations obtained from a pre-trained policy. While this expert policy generally performs the task successfully, it is not necessarily optimal. Limitations in the expert's strategy, such as suboptimal maneuvering, local minima in control, or conservative safety heuristics, can result in trajectories that are effective but not globally optimal. Understanding these imperfections is crucial, as they define the upper bound for pure imitation learning and provide context for methods that aim to improve upon the expert behavior.

The sample efficiency evaluation demonstrates that all algorithms benefit from additional demonstrations, with gains diminishing beyond a certain number of trajectories. Interestingly, TD3-BC consistently outperforms IQL at nearly all data points, and in some cases even exceeds the expert policy when trained on the maximum number of demonstrations. This can be explained by the fact that these methods optimize a policy based on the observed trajectories while also leveraging the reward signal, allowing them to refine and improve upon suboptimal behaviors in the expert data.

Table 1: Mean final returns for BC policies with varying network sizes (neurons per hidden layer).

Network size	Mean final return
16	332.8
32	334.5
64	342.8
128	355.6
256	370.9

At very low demonstration counts, offline RL methods show slightly better performance than behavioral cloning, likely because BC is purely supervised and struggles to generalize from extremely limited data, whereas offline RL can exploit even sparse feedback to guide learning.

The robustness experiment indicates that observation noise degrades performance across all algorithms, although some methods exhibit greater resilience. The model capacity ablation confirms that smaller networks struggle to capture expert behavior, while larger networks achieve higher returns. The most notable improvements are observed when increasing from 64 to 128 neurons and from 128 to 256 neurons, with the latter yielding the largest single gain. While performance increases consistently with model size, the results suggest that returns may eventually plateau beyond the tested range.

Overall, there are no significant differences between the algorithms when sufficient demonstrations are available. This suggests that for this task, the combination of task simplicity, deterministic dynamics, and rich expert trajectories allows all methods to reach near-optimal performance, with differences emerging mainly in low-data or noisy conditions.

5 Conclusions and future work

In this work, we demonstrated successful model-free learning of the tractor-trailer task using reinforcement learning. The trained expert policies provide a solid foundation for imitation learning, which we explored in an offline setting. Our results show that imitation learning can be effectively performed using both pure Behavioral Cloning and offline RL methods, with all approaches achieving near-optimal performance when sufficient demonstrations are available.

For future work, we plan to extend these experiments to more realistic, high-fidelity simulations that better capture the variability of real agricultural environments, including uneven terrain, varying soil conditions, and sensor noise. Ultimately, we aim to validate the learned policies on a physical tractor-trailer system, which will require careful consideration of safety, stability, and robustness. Since this work focuses on simulation-based learning, it serves as a proof-of-concept demonstrating the efficacy of

imitation learning methods for trailer control. Field experiments constitute the natural next step in this research program. While the current study focused on offline imitation learning, interactive or online learning schemes could further improve performance and adaptability, provided that safety constraints and reliable policy updates are ensured.

Furthermore, the presented approaches can potentially be adapted for edge-based processing. Additionally, the current formulation does not explicitly address obstacle avoidance; however, the reward-based framework can be readily extended to incorporate collision penalties or safety constraints, enabling deployment in more complex real-world environments. Achieving this would require considerations such as computational efficiency, sensor integration, and low-latency inference, which are important steps towards deploying these learning algorithms in practical, real-time applications.

Overall, this work highlights the flexibility of both imitation learning and reinforcement learning methods, and provides a clear path towards interactive and real-world implementations.

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Bionotes



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